Attention-Driven Hybrid RAG for Scientific Discovery

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ABSTRACT
Large language models often suffer from hallucinations in specialized scientific literature. In this paper, we propo

1. INTRODUCTION
Information retrieval over scientific papers requires both exact lexical keyword matching for technical terminology

2. METHODOLOGY

Our proposed pipeline operates in three stages:

1. Lexical Token Indexing: We build an inverted BM25 index over sliding-window text chunks of 512 tokens with
2. Dense Embedding Vectorization: Chunks are projected using MiniLM-L6-v2 embeddings into a normalized cos
3. Hybrid Score Fusion & Cross-Encoder Reranking: Reciprocal rank fusion combines the lexical and semantic c

3. EXPERIMENTS AND RESULTS

We benchmarked our system across two academic QA datasets: PubMedQA and SQuAD_v2.

- Baseline Dense (SentenceTransformer): Accuracy = 0.812, F1 = 0.845
- Lexical Only (BM25): Accuracy = 0.785, F1 = 0.801
- Proposed Hybrid RAG: Accuracy = 0.891, F1 = 0.924

The hybrid architecture demonstrates a +7.9% F1 improvement on scientific question answering.

4. DISCUSSION AND LIMITATIONS

While the proposed hybrid pipeline demonstrates significant gains, several limitations remain.
First, cross-encoder reranking adds 120ms of inference latency per query on CPU compute, requiring GPU acce
Second, the system has not yet been tested on multimodal papers containing intricate mathematical figures and
Third, long-context documents exceeding 100 pages require hierarchical chunking to avoid context dilution.

5. CONCLUSION AND FUTURE WORK

We have presented Attention-Driven Hybrid RAG, a robust framework for grounded question answering over sch